VVU · COMPREHENSIVE PLANNING FRAMEWORK

Venture Vision Ubuntu

The Complete Estate, Compiled for the Quad-Partite Alliance, the Department of Water and Sanitation, and the Local Municipality

[LIVE] • EDGE: GQEBERHA • WORM HEAD: **E965B01D**

This document compiles the entire estate — the architecture, the testimony, the kernel, the water hub, the audit, the platform, the go-live, the twelve clusters, the evidence architecture, and the two codebases — into a single navigable framework that a partner can read, verify, and act on. It is the one document the founder hands to every partner: the Quad-Partite Alliance, the Propella/PIA acceleration partners, the Department of Water and Sanitation, and the local municipality. Every claim in this document carries a tier tag. Every number is measured. Every source is named. The estate proves the record, not the reality. The fall is part of the design.

VAGUELY VANITY LLC (PTY) LTD T/A VENTURE VISION UBUNTU
14 BIRD STREET, GQEBERHA · FILED 2026-10-15

CIPC 2026/259053/07 · SARS 9106703292 · WORM HEAD E965B01D

Entity details per B-BBEE Certificate 945551886 (CIPC, 26 March 2026). Canonical source: the certificate, not this document.

The Estate

Section 1 · the founder, the company, the registrations, and the cryptographic anchor — the estate's identity, filed once, verified by anyone.

SECTION 1 OF 12 · THE ESTATE

1.1 The Three Doors

The estate is built on a doctrine of separation that is older than the company that holds it. Three legal persons, three technical layers, one shared ledger. **Door 1 — Legal:** Vaguely Vanity LLC (Pty) Ltd, the registered entity that files tax, holds the CIPC registration **2026/259053/07**, carries legal liabilities, and operates the Capitec Business Banking account. **Door 2 — Infrastructure:** Venture Vision Ubuntu, the trading name that governs the off-grid water hubs, the edge mesh deployments, and the WORM chain head **E965B01D**. **Door 3 — Editorial:** Vaguely Magazine, the citable public journal and IEEE-style archive. The three doors are firewalls. The doors do not touch each other directly; they touch only the ledger, and the ledger touches nothing back. The separation is the architecture's refusal to let a failure in one layer travel into another.

The three doors are the estate's first answer to the question a partner asks: *who am I dealing with, and how do I know the entity is real?* The answer is three legal persons, each verifiable against a South African state registry, and a cryptographic anchor that ties the three to a single immutable chain. The partner who reads this document is reading a thing published by Door 3 (the magazine), about a platform operated by Door 2 (the infrastructure), held by Door 1 (the legal entity). The three are separate. The three are honest. The separation is the honesty.

1.2 The Entity

The entity is **Vaguely Vanity LLC (Pty) Ltd t/a Venture Vision Ubuntu**. The registration is **CIPC 2026/259053/07**. The SARS tax reference is **9106703292**. The B-BBEE rating is Level 1 (135% Preferential Procurement Recognition). The banking is Capitec Business, active. The domicile is 14 Bird Street, Central, Gqeberha, 6001, Eastern Cape, South Africa. The entity details are per B-BBEE Certificate 945551886, issued by CIPC on behalf of the dtic on 26 March 2026. The certificate is the canonical source. The certificate, not this document, is the thing a partner verifies against. The document is the route to the certificate. The certificate is the receipt.

The entity has no (RF) suffix. The registered name, per the B-BBEE certificate, is VAGUELY VANITY LLC (PTY) LTD, trading as Venture Vision Ubuntu. The (RF) variant that appears in older documents is a thing the estate has corrected, and the correction is the thing the checklist filed on 2026-10-12: zero instances of (RF) in the canonical base. The correction is the estate's refusal to carry a suffix the certificate does not carry, and the refusal is the same discipline the Zero Fabrication Rule enforces: the system may not manufacture a name it does not have.

1.3 The WORM Head

The WORM head is **E965B01D**. The eight characters are the short form of a SHA-256 digest, and the digest is the Merkle root of the first batch of telemetry the estate ever folded. The head is the receipt for the whole estate, the way the commit hash is the receipt for the code. A partner who reads **E965B01D** on the cover of this document and reads **E965B01D** on the chain knows, without trusting any voice in this volume, that the document and the chain are talking about the same ledger. The head is the cross-reference. The head is the anchor. The head is the thing the next eleven sections unfold.

The Architecture

Section 2 · the Skylantern primitive — four layers, three levels, six primitives, one funnel — the general form of the estate's evidence-verification architecture.

SECTION 2 OF 12 · THE ARCHITECTURE

2.1 The Four-Layer Stack

The architecture is a four-layer stack, and the stack is the thing the estate calls the Skylantern primitive. **Layer 1 — Physical Device:** the sensor, the dongle, the phone, the meter — the device that sits in the pipe, in the car, in the pocket, and reports what is there. **Layer 2 — Hardware Crypto & Signing:** the layer that takes the raw reading, stamps it with an ISO timestamp, hashes it with SHA-256, signs it with the hardware key, and hands the signed payload to the next layer. **Layer 3 — WORM Chain Log:** the append-only sequence of signed payloads, each linked to the one before by a hash, the whole sequence anchored to **E965B01D**. **Layer 4 — User-Facing App:** the dashboard, the alert, the report — the layer that displays the receipt to the person who needs to act on it. The four layers are the master primitive. **One master primitive. Infinite realities.** Every Skylantern product is merely a reconfiguration of this exact four-layer stack.

2.2 The Three Levels

The architecture refuses to collapse three levels into one, and the refusal is the key rule. **Step 1 — Observed Fact:** what the sensor reported. The only absolute truth the system holds. A pressure drop of 1.2 bar over 4 minutes. The number. Not the interpretation. **Step 2 — Derived Event:** what the software inferred. Leak-candidate flagged. Always labelled with the firmware version that made the inference. **Step 3 — Decision:** what action the system recommended or took. Maintenance alert SMS dispatched. The three levels are a chain of causation — fact → inference → decision — and the chain is the thing the architecture refuses to collapse, because a collapsed chain is a chain where the decision is mistaken for the fact, and a decision mistaken for a fact is the thing the WORM ledger was built to refuse. The raw observation must always remain recoverable.

2.3 The Six Primitives

The funnel narrows 126 sensor types across 15 device classes to 6 functional primitives, and the 6 are the questions every Skylantern product is built from. **Motion** — is the thing moving? **Proximity** — is the thing near another thing? **State** — is the thing on, off, open, closed? **Identity** — is the thing the thing it claims to be? **Location** — where is the thing? **Environment** — what surrounds the thing? The six are not sensor types. The six are questions, and the questions are the thing the 126 sensors are translated into. A flow meter answers Motion and State. A BLE beacon answers Proximity and Identity. An OBD-II dongle answers State and Motion. The six questions cover the 126 sensors.

2.4 The Standardization Funnel

The funnel is **ruthless**, and the ruthlessness is the paranoia. A generous funnel keeps the 126 sensor types as 126 separate vocabularies, and 126 separate vocabularies is 126 places where a bug can hide, a drift can go unfiled, an inference can be collapsed into an observation. The ruthless funnel collapses the 126 into 6, and the 6 are small enough to audit. The funnel is the estate's Three Doors applied to sensor types: keep the set small, keep the set legible, keep the set explainable on the worst day of the worst hearing. The funnel is the thing that makes the primitive a primitive rather than a catalogue.

The Kernel

Section 3 · CircuitBreaker.sol — the 26-line Solidity kernel, the 11-Beat Loop, the 32-byte memory wipe, and the 24-hour disconnect survival.

SECTION 3 OF 12 · THE KERNEL

3.1 CircuitBreaker.sol

The kernel is twenty-six lines of Solidity, and the twenty-six are the thing the founder wrote in a single night in July 2026, after a secondary hearing flared over disputed audio recordings and an unverified chain of custody. The contract is named **CircuitBreaker** because that is what it is — a breaker that trips the moment legal truth becomes uncertain, and that, once tripped, refuses to un-trip. The contract has no constructor, no owner, no upgrade proxy, no governance token, no fee mechanism, no pauser role. The featurelessness is the feature. Every line that is not in the contract is a line that cannot be exploited, cannot be social-engineered, and cannot be quietly reinterpreted by a future maintainer.

```
// SPDX-License-Identifier: MIT
// CircuitBreaker.sol: Write Once, Read Many (WORM) Ledger Structure
pragma solidity ^0.8.20;
contract CircuitBreaker {
    event InclusionProof(bytes32 indexed root, bytes32 indexed receipt);
    struct WORMLedger {
        bytes32[] historicalRoots;
        uint256 currentIndex;
        bool isFrozen;
    }
    WORMLedger public ledger;
    function submitRoot(bytes32 _root) external {
        require(!ledger.isFrozen, "Ledger is frozen: Legal truth uncertain");
        ledger.historicalRoots.push(_root);
        ledger.currentIndex++;
        emit InclusionProof(_root, sha256(abi.encodePacked(_root, ledger.currentIndex)));
    }
    function freeze() external {
        require(ledger.currentIndex > 0, "Cannot freeze empty ledger");
        ledger.isFrozen = true; // Fail-closed safety floor
    }
}
```

The revert string is the kernel's only sentence, and it is a sentence a court can read: **"Ledger is frozen: Legal truth uncertain"**. The string is the contract's refusal to be only a piece of code — it is the contract's insistence that it is also a document, a thing a person can read in a room without compiling it. The string is the kernel's testimony, and the testimony is the thing the WORM head **E965B01D** anchors.

3.2 The 11-Beat Loop

The kernel does not ingest telemetry directly. The kernel ingests roots — the SHA-256 digests that stand in for batches of telemetry. The thing that produces those roots is the **11-Beat Loop**: the cycle an edge node runs every time it ingests a telemetry frame. The eleven beats are: **Ingest → Validate → Sign → Buffer → Fold → Root → Submit → Replicate → Verify → Audit → Seal**. Each beat is a checkpoint. Each checkpoint is a

refusal of the idea that a record can be trusted before it has been verified at the stage it was made. The loop produces the root; the kernel seals it; the seal is the receipt; the receipt is the record.

3.3 The 32-Byte Memory Wipe

Each cycle of the 11-Beat Loop ends with a memory wipe. After the seal, the node's working memory is wiped down to **32 bytes** flat — the size of a single **bytes32**, the root of the cycle just sealed. Everything else — the frames, the signatures, the intermediate folds — is cleared. The wipe is the discipline that lets an ESP32-S3 run the loop continuously, for months, without buffer overflows or state corruption. The kernel is twenty-six lines because every line that is not in the contract is a line that cannot be exploited. The memory is thirty-two bytes because every byte that is not in the memory is a byte that cannot be corrupted. The two disciplines are the same discipline, at two scales.

3.4 Edge Resilience

The edge node survives a twenty-four-hour network disconnect by buffering signed frames in a local **SQLite** database and submitting the buffered roots to the kernel with exponential backoff upon reconnection. The first retry waits a second; the second waits two; the third waits four; the fourth waits eight. The 24-hour disconnect, the SQLite buffer, and the exponential backoff are the three specifications that let the edge node survive the network the township actually has — a network that drops, that drops often, and that drops for hours at a time. The grid, when it is there, is a convenience. The grid, when it is not, is not a dependency.

The Water Hub

Section 4 · the 12-metre container, the Kit of Parts (no cash requested), the Hydro-Bayesian Kernel Mk-II, and the governing principle: the AI never overrides physics.

4.1 The 12m Container

A 12-metre shipping container is not a beautiful object, and the estate has chosen it anyway. The container is the most abundant, the most repurposable, and the most legible piece of civic hardware the Eastern Cape has. It is the thing a municipality knows how to deliver, a sponsor knows how to donate, and a community knows how to receive. The hub is built inside a container because the container is the language the township already speaks. The container is the unit of the hub's replicability. The unit is the replicability. The replicability is the only honest answer to a water question that is not, finally, a question for one hub.

4.2 The Kit of Parts

The hub is built from a Kit of Parts, and the Kit of Parts is built on a single discipline: **no cash requested**. The estate does not ask a sponsor for money. The estate asks a sponsor for materials. Four line items, four sponsors, four SDG targets. **12m Shipping Container** — 1 Unit — Transnet / Logistics Sponsor — **SDG 9 · Industry & Infrastructure**. **5,000L Jojo Water Tank** — 1 Unit — Local Hardware Supplier — **SDG 6 · Clean Water & Sanitation**. **100W Solar Panel + Battery** — 1 Array — Renewable Energy Sponsor — **SDG 7 · Affordable & Clean Energy**. **ESP32-S3 Sensors & Bench** — 1 Rig — Engineering / Tech Sponsor — **SDG 4 · Quality Education**. The four materials are the hub. The hub is not a favour. The hub is four materials, four sponsors, four receipts. The receipts are the load-bearing wall.

4.3 The Hydro-Bayesian Kernel Mk-II

The Hydro-Bayesian Kernel Mk-II — **HBK Mk-II** — is the stainless-steel pipe rig that does the hub's actual sensing. The rig is built from **316L/304** stainless steel piping with **15.0 mm** isolation gaps. The two grades of steel are specified because they are the two grades a South African stainless-steel supplier is most likely to stock. The 15.0 mm gap is the distance the rig keeps between adjacent pipe runs, and the distance is the most carefully chosen number in the hub's geometry — the gap that lets the acoustic sensors distinguish a leak in one pipe from a vibration in the next. The number is the number the founder measured, not the number the founder estimated.

4.4 The AI Never Overrides Physics

[DOCTRINE] HBK MK-II · GOVERNING PRINCIPLE

The AI never overrides physics; if physical sensor readings contradict model inference, the system defaults to raw physics.

The principle is the base layer's constitution. The AI — the particle filter, the cloud of plausible truths — is a guest in the pipe, not the host. The sensor is the host. The principle is the rule that, when the two disagree, the sensor wins, and the AI's output is set aside and the raw sensor reading is the reading the hub records. The principle is the founder's refusal to let the hub repeat, in steel, the institutional error he met in 2018 — the error of applying a model to a physics and editing the physics out. The sensor is the witness. The AI is the interpreter. The two are not the same, and the principle keeps them apart.

The Evidence Architecture

Section 5 · the Oracle Boundary, the Zero Fabrication Rule, the Corroboration Line, and the Commercial Constitution — the estate's confession of what it can do and what it refuses to claim.

SECTION 5 OF 12 · THE EVIDENCE ARCHITECTURE

5.1 The Oracle Boundary

The Oracle Boundary is the line between the physical world and the immutable record, and the line is labelled with a warning the architecture has refused to remove: **A chain proves the record was not altered. It does not prove the reading was true.** The sentence has two clauses. The first is the promise. The second is the limit. A chain that proved the reading was true would be a chain that was lying, because a chain is mathematics, and mathematics cannot know whether a thermistor was calibrated. The chain can know only what the chain was given — a hash of a payload signed by a key. The hash proves the payload has not changed. The key proves the payload came from the device. Neither proves the payload was a true reading of the world. The warning is the honesty. The honesty is the boundary.

5.2 The Zero Fabrication Rule

The Zero Fabrication Rule is the architectural sibling of the WORM ledger's fail-closed safety floor. **Once the hardware key signs the payload, silent deletion becomes mathematically impossible for both the user and the operator. Missing data is never guessed — it is flagged as UNDEFINED. The system may not manufacture evidence.** The rule has two halves. The first — silent deletion is impossible — is enforced by the hash and the key. The second — missing data is never guessed — is enforced by the firmware, which is required to flag a missing reading as **UNDEFINED** rather than interpolate, average, or impute. The rule is the estate's refusal to be the institution that quietly manufactures evidence. The rule is the receipt for the honesty.

5.3 The Corroboration Line

A record arrives at the chain as a **CANDIDATE** — the single-witness state. The chain holds the record but does not endorse it. The record elevates to **VERIFIED** only when a second witness is attached — a paired wearable device, an external GNSS anchor, or a second mobile device. The promotion is not a thing the system does on its own; the promotion is a thing the second witness does, by signing, by timestamping, by reporting from a different position. **Court-grade eligibility begins only at Tier R2.** No record is silently upgraded. If there is no second witness, the chain explicitly states the lower evidentiary weight. If verification fails, the chain does not hide the failure — it labels the record **TAMPER_DETECTED**. The corroboration line is the WORM ledger's fail-closed discipline applied to evidence.

5.4 The Commercial Constitution

The constitution is a band, and the band is **R9 to R9,999**. The cap is a design constraint, not a marketing decision. The cap forces the architecture to build inside the band, and the forcing is the thing that keeps the product small, honest, and affordable. A product that costs R50,000 is a product that needs a procurement committee, and a procurement committee is the institution the founder has spent the decade refusing. **If a product cannot be built inside this band, it is not a Skylantern product.** The enforcement clause is the commercial version of the Zero Fabrication Rule: the system may not manufacture a reading, and the architecture may not manufacture a price. A fabricated price is the commercial equivalent of a fabricated reading.

The Platform

Section 6 · the go-live, the dashboard, the Upload Dock, and the revenue share — the estate as a publishing platform, not just a magazine.

SECTION 6 OF 12 · THE PLATFORM

6.1 The Go-Live

The platform is live. The go-live evidence was filed on **2026-09-15** at 22:49:36 UTC, commit **cd7d785**, runner **tests/go-live/run-golive.ts**, all checks executed in one pass, none copied. The numbers the checklist filed on 2026-10-12: **106 routes**, **38 Prisma models**, **472 source files**, **6 mini-services**, lint clean, zero errors. The MCP Orchestrator Suite: **4/4 gates PASS**, **5 claims UNVERIFIED**. The asymmetry is the discipline: a gate that passes is a gate the architecture has verified; a claim that is unverified is a claim the architecture has refused to promote. Two warnings flagged honestly: telemetry **:3003** DISABLED (4GB memory limit), instrumentation workers DISABLED. A checklist that flags two failures is a checklist that is telling the truth.

6.2 The Dashboard

The dashboard is the estate's live webpage — a media-first, paranoia-first landing page that strips away 80% of the text and replaces it with visual anchors: a full-screen video hero, an inline audio player, an interactive timeline, a 3D render placeholder, and prominent QR codes. The dashboard carries a telemetry strip — Trust Score **70.01%**, Chain Head **E965B01D**, System State **STEADY_STATE_LOCKED**, Real Students **8** — four numbers, four receipts, in a single bar across the page. The numbers are the estate's first answer to the question a reader asks: does the estate's record match the estate's marketing? The answer is the telemetry strip, and the strip is the thing the next two subsections extend.

6.3 The Upload Dock

The Upload Dock is the estate's Spotify moment — the open gate that turns a library into a platform. The dock is open to four kinds of shipper, unranked: **LECTURERS**, **ARTISTS**, **MUSICIANS**, **ORDINARY PEOPLE**. A lecturer's article and a grandmother's story arrive at the same gate, are filed into the same ledger, and are anchored to the same WORM head. The dock carries an indemnity and takedown regime: the shipper warrants they hold rights; the estate operates notice-and-takedown; the shipper indemnifies the estate against third-party IP claims. The regime is the load-bearing wall beneath the open gate — the thing that keeps Spotify-for-Publication plus verbatim IEEE-style content from making the estate the defendant.

6.4 The Revenue Share

The dock offers three frees — **Free Publishing**, **Free Hosting**, **Free Distribution** — and takes a split: **30% platform fee / 70% creator share**, calculated on **net revenue** (gross revenue less payment-processing costs and third-party platform fees, before any

recoupment of opt-in advances). The creator retains ownership of the masters. The estate anchors, hosts, and distributes; the estate does not own. Recoupment is opt-in and separately documented. Payouts are quarterly, with a three-to-six-month lag. The split, the net-revenue definition, the opt-in recoupment, the lag, and the quarterly payout are the five terms of the bargain, filed per submission in a one-page agreement anchored to **E965B01D**. The revenue share is the commercial version of the WORM ledger: the ledger refuses to let the record be quietly rewritten, and the revenue share refuses to let the creator be quietly unpaid.

The HBK Mk-II Dashboard

Section 7 · the actual codebase — 38 Prisma models, 22 API routes, the WORMEvent model, and the CityTwin3D spatial telemetry nodes — the platform's data architecture, filed in code.

SECTION 7 OF 12 · THE HBK MK-II DASHBOARD

7.1 The Prisma Schema

The HBK Mk-II dashboard is built on a Prisma schema with 38 models, and the 38 are the thing the platform's data architecture is made of. The schema is the estate's data constitution — every model carries a tier tag, every status field is enumerated, every relationship is explicit. The key models the partner needs to know:

```
model WORMEvent {
  id              String   @id @default(cuid())
  hash            String   @unique
  previousHash    String?
  wormHead        String
  subject         String
  source          String
  tier            Int
  verificationState String  @default("INCONCLUSIVE")
  // INCONCLUSIVE | CANDIDATE | VERIFIED | TAMPER_DETECTED
  payload         String   // JSON blob of the evidence payload
  observedAt      DateTime @default(now())
}

model TelemetryNode {
  id              String   @id @default(cuid())
  nodeId          String   @unique // e.g. ESP32-S3-HUB-01
  type            String   // edge_node | tank | pump | valve | acoustic
  lng             Float
  lat             Float
  wormHead        String?
  verificationState String  @default("INCONCLUSIVE")
}
```

The WORMEvent model is the chain's data structure: every event carries a hash, a previousHash, a wormHead, a tier, and a verificationState. The verificationState field is the thing the Corroboration Line reads — **INCONCLUSIVE**, **CANDIDATE**, **VERIFIED**, or **TAMPER_DETECTED**. The four states are the four things the chain refuses to collapse, the way the three levels are the three things the key rule refuses to collapse. The TelemetryNode model is the spatial layer — each node has a type (edge_node, tank, pump, valve, acoustic), a longitude, a latitude, and its own wormHead. The node is the thing the CityTwin3D map pins.

7.2 The API Routes

The dashboard exposes 22 API routes, and the 22 are the thing the platform serves. The routes the partner needs to know: **/api/worm** and **/api/worm-chain** — the WORM ledger endpoints, the thing that returns the chain's validity, receipt count, and merkle root. **/api/telemetry** — the telemetry endpoint, the thing that returns the live sensor readings. **/api/gates** — the MCP gate endpoints, the thing that returns the 4/4 PASS and 5 UNVERIFIED. **/api/deliverables** — the deliverables endpoint, the thing that returns the

R0-R3 and C1-C3 phase deliverables, each with a tier tag and an acceptance criterion. **/api/onboard/consent** — the consent endpoint, the thing that files the Section 69 consent gate. The 22 routes are the platform's surface, and the surface is the thing a partner integrates against.

7.3 The WORM Event Model

The WORMEvent model is the thing that makes the chain a chain. Every event has a **hash** (SHA-256 of the payload plus the previous hash), a **previousHash** (the hash of the event before it), and a **wormHead** (the anchor, **E965B01D**). The three fields together are the thing that makes the record tamper-evident: alter one event, and every hash that follows breaks. The **verificationState** field is the thing the Corroboration Line writes to: a CANDIDATE that gets a second witness is promoted to VERIFIED; a CANDIDATE whose second witness contradicts is labelled TAMPER_DETECTED. The model is the Skylantern primitive's third layer, instantiated in Prisma. The model is the receipt. The model is the chain. The model is the thing the next section's visual state contract renders.

7.4 The CityTwin3D Spatial Telemetry

The TelemetryNode model is the thing the CityTwin3D map renders — each node is a pin on the map, with a type, a position, and a verification state. The CityTwin3D is the estate's spatial intelligence layer — the thing that lets a municipal engineer see every sensor, every tank, every pump, every valve, on a map of the network, with the verification state of each node colour-coded. The spatial layer is the thing the Anatomy integration (next section) renders visually, and the rendering is the thing that turns the Prisma model from a data structure into a thing a person can see and act on. The node is the pin. The pin is the receipt. The map is the thing the municipality reads.

The Anatomy Integration

Section 8 · the Anatomy×VVU integration — the typed visual-state contract, the adapter pattern, the evidence instrument, and the acceptance tests that keep the figure from fabricating verification.

SECTION 8 OF 12 · THE ANATOMY INTEGRATION

8.1 The Visual State Contract

The Anatomy integration is built on a typed visual-state contract, and the contract is the thing that keeps the figure from fabricating verification. The contract defines an **AuthoritativeStatus** with six values: **READY**, **BLOCKED_MISSING_INPUT**, **INCONCLUSIVE**, **PROVEN**, **FAILED**, **UNKNOWN**. The figure renders the status; the figure does not compute it. The contract defines a **VisualState** with the entity ID, the authoritative status, the status reason, the data mode (live, replay, simulation), the observed-at timestamp, the source status, the values, the missing inputs, and the evidence references. The contract is the thing the adapter produces, and the adapter is the thing the next subsection describes.

```
export type AuthoritativeStatus =
  | "READY"
  | "BLOCKED_MISSING_INPUT"
  | "INCONCLUSIVE"
  | "PROVEN"
  | "FAILED"
  | "UNKNOWN";

export interface VisualState {
  entityId: string;
  entityLabel: string;
  authoritativeStatus: AuthoritativeStatus;
  statusReason: string;
  mode: DataMode; // live | replay | simulation
  observedAt?: string; // ISO 8601
  sourceStatus: 'available' | 'stale' | 'missing' | 'unknown';
  values: Record<string, { value: number | string | boolean; unit?: string }>;
  missingInputs: string[];
  evidence: EvidenceReference[];
}
```

8.2 The Adapter Pattern

The adapter is the thing that maps the authoritative state to the visual state, and the mapping is the thing the acceptance tests check. The adapter does not infer missing evidence. The adapter does not recompute proof. The adapter maps statuses and values, and the mapping is a pure function — the same input always produces the same output, and the output is the thing the figure renders. The responsibility boundary is the thing the integration plan files: verification rules and results are owned by VVU·IVE; evidence and provenance are owned by ProofBridge; spatial coordinates are owned by the spatial application; sensor telemetry is owned by the edge ingestion service; figure geometry and animation are owned by Anatomy; routes, identity, and tenant access are owned by the existing dashboard. The boundary is the thing that keeps the figure from becoming the verifier.

8.3 The Evidence Instrument

The evidence instrument is the thing the integration produces — a visual figure that renders the authoritative state and emits typed intent events when a user interacts with it. The events are three: **open-evidence** (the user clicked a link to an evidence record), **select-component** (the user selected a part of the figure), **request-refresh** (the user asked for a fresh state snapshot). The three events are the only things the figure emits, and the three are the thing the dashboard handles. **Intentionally, no event exists for setting status to PROVEN or releasing a product.** The sentence is the contract's enforcement clause, and the clause is the thing that keeps a click, an animation, or a colour change from altering the authoritative verification result. The figure displays. The figure does not decide.

8.4 The Acceptance Tests

The acceptance tests are the thing the integration must pass before production, and the tests are the thing the partner can check. State fidelity: each authoritative status maps to exactly one defined display state; **BLOCKED_MISSING_INPUT** never appears as **PROVEN**; **INCONCLUSIVE** remains distinct from failure and success; a render error does not alter the authoritative state. Evidence and provenance: every evidence link uses an ID supplied by the authoritative service; missing provenance is shown as missing; the renderer does not fill gaps; the figure does not claim cryptographic verification unless the service's verified record supports that claim. Interaction and accessibility: all actionable components are keyboard-reachable; state is not communicated by colour alone; a readable text summary exists without animation. Performance: initial render and update cost measured on a representative mobile device. The tests are the thing that turns the integration from a prototype into a production-grade visual evidence layer.

The Radical Honesty

Section 9 · the August 20 audit, the twelve editorial clusters, the four hosted articles, and the five-minute audit anyone can run.

9.1 The August 20 Audit

On August 20, 2026, the founder filed an unvarnished internal audit — **VVU-AUD-20260820** — confronting the estate's marketing claims against its codebase reality. Four rows. Four marketing claims. Four codebase realities. Four statuses. Three **[FILED FINDING]** and one **[MEASURED (8)]**. Database integration: marketing claimed a complete PostgreSQL bridge; reality: zero database integration, static in-memory mocks. Authentication: marketing claimed hardened multi-tenant auth; reality: cosmetic frontend authentication. User interface: marketing claimed six live production panels; reality: six partial UI sections. Enrolled students: marketing claimed enterprise deployment active; reality: 8 students enrolled, 0 farmers on the mesh. The audit is the estate's testimony against itself, filed in the same volume as its testimony for itself. The audit is the product.

9.2 The Twelve Clusters

The twelve clusters are the platform's editorial voice, drafted at **r50** through **r53** on 2026-09-30, from the estate's own code and ledger. Three clusters, four articles each. **Cluster 1 — The Receipt:** the pipe that talks, the notebook that cannot be quietly edited, what the numbers mean, check it yourself. **Cluster 2 — The Law:** the paper the law asks for, consent you can take back, keep only what you can stand behind, when it breaks say so. **Cluster 3 — The Audit:** the audit is coming, prove it hasn't changed, the receipts people actually keep, the five-minute audit. The twelve are the estate's refusal to let the platform be a thing that only engineers can read.

9.3 The Four Articles

The four Asterisk articles are the platform's first hosted content — the thing the Upload Dock was built to open. *So You Think You Could Be An Electrician?* by Jesse Smith — the trades vs. college question, complicated. *What To Expect When You're Expecting (A World of Multi-Agent Systems)* — the AI age, framed. *What is a Super El Niño?* from NPR — the climate, charted. *After Work, We'll Have Each Other* — the future of labour, asked. The four are hosted, not owned. The estate anchors them to the WORM head, files them in the Media Library, and links back to the originals. The creator retains the masters. The estate takes thirty percent of the net; the seventy percent is the creator's, quarterly, with a receipt. The four articles are the platform as a venue, not a publisher.

9.4 The Five-Minute Audit

The five-minute audit is the thing that closes the radical honesty, and the thing is a checklist a person with a terminal can run. **Five minutes, four checks.** First: is the

chain valid? Second: what is the receipt count? Third: what is the merkle root, and does it match the anchor? Fourth: what is the WORM head, and is it **E965B01D**? The four checks are the thing the estate has refused to make complicated, because an audit that takes five minutes is an audit that happens, and an audit that happens is the thing the estate was built to survive. The five-minute audit is the thing a partner runs before signing, the thing a municipality runs before deploying, and the thing a sceptical ratepayer runs before trusting. The audit is the receipt for the receipt.

The Partnership Ask

Section 10 · what each partner is asked for — the Quad-Partite Alliance (materials, not cash), the DWS (domain validation), the municipality (institutional support), and Propella/PIA (acceleration).

10.1 The Quad-Partite Alliance

The Quad-Partite Alliance is the four sponsors the Kit of Parts requires, and the ask is **materials, not cash**. **Transnet / Logistics Sponsor** — the 12-metre shipping container, one unit, SDG 9. **Local Hardware Supplier** — the 5,000-litre Jojo water tank, one unit, SDG 6. **Renewable Energy Sponsor** — the 100W solar panel and LiFePO4 battery, one array, SDG 7. **Engineering / Tech Sponsor** — the ESP32-S3 sensors and bench, one rig, SDG 4. The four sponsors are the four doors of the Kit of Parts, and the four are the thing the estate files against named SDG targets. The ask is the smallest correct thing: four materials, four sponsors, four receipts. The receipts are the load-bearing wall.

10.2 The DWS 72-Hour Validation

The Department of Water and Sanitation is asked for one thing: **domain validation**. Does the engineering model VVU has encoded map correctly to how DWS and municipalities actually observe, diagnose, and verify infrastructure problems? The ask is the 72-hour validation protocol: DWS provides governed historical or test data, the primitive runs against the data for 72 hours, and the 72 hours measure whether the primitive identifies the patterns DWS engineers already recognise. The protocol is not a demo. The protocol is a test. If the answer is yes, the primitive is VERIFIED — elevated, by the DWS as the second witness, from CANDIDATE to the thing a court may consider. If the answer is no, the primitive is returned to the bench. The estate has refused to fear the return, because a primitive tested honestly is the thing the Zero Fabrication Rule demands.

10.3 The Municipality

The local municipality is asked for two things: a site for the 12-metre container, and the institutional support that lets the hub operate beside a school or a clinic. The site is the thing the container sits on — a patch of compacted earth, a corner of a schoolyard, a space beside a community centre. The institutional support is the thing that lets the hub connect to the municipal water network (for the bench to listen to) and to the municipal power grid (as a backup to the 100W solar). The hub is off-grid by design, but the grid, when it is there, is a convenience, and the convenience is the thing the municipality provides. The ask is the site and the support. The hub is the thing the municipality gets.

10.4 Propella and PIA

The Propella and PIA acceleration partners are asked for two things: acceleration support (the PreSeed funding the TIA/DSBD application seeks) and the institutional

mentorship that turns a platform from a thing that is live into a thing that is sustainable. The TIA/DSBD application has been filed — the thing the estate submitted for the Technology Innovation Agency and the Department of Small Business Development, the agencies whose PreSeed funding the platform is seeking. The Propella proposal has been filed — the thing the estate files for the accelerator that asks the estate to explain itself in the language a reviewer expects. The two asks are the estate's first institutional filings, and the filings are the things the estate has filed in writing, before the answer, with the radical honesty the estate was built to enforce.

The Roadmap

Section 11 · the path from Tier 2 (Specified) to Tier 3 (Verified), the 72-hour protocol, the ten-year horizon, and Issue 2 ("Flow").

SECTION 11 OF 12 · THE ROADMAP

11.1 Tier 2 (Specified) to Tier 3 (Verified)

The estate is filed at **Tier 2 (Specified)** — the tier of a thing that has been designed, documented, and filed but not yet validated against the world it claims to observe. Tier 2 is the CANDIDATE tier. The 72-hour validation protocol is the thing that, if it succeeds, elevates the status to Tier 3 — the VERIFIED tier, the court-grade tier, the tier where the second witness has been attached and the record has been measured against the reality. The estate has refused to claim a tier it has not earned, and the earning is the thing the DWS engineers are asked to provide. The roadmap is the path from the specified to the verified, and the path is the thing the Zero Fabrication Rule files: nothing is presented as operational unless it is.

11.2 The 72-Hour Protocol

The 72-hour protocol is the single test the estate has asked the DWS to run, and the test is the thing that either confirms or returns the primitive. DWS provides governed data. The primitive runs against the data for 72 hours. The 72 hours measure whether the primitive identifies the patterns DWS engineers already recognise. If yes: VERIFIED, Tier 3, court-grade. If no: returned to the bench, re-specified, re-tested. The protocol is the estate's refusal to ask the DWS to take the primitive on faith. The protocol is the thing that lets the DWS engineers see the primitive work against their own data, in their own time, with their own patterns, before anyone asks them to trust it. The 72 hours are the corroboration line applied to the submission. The DWS is the second witness.

11.3 The Ten-Year Horizon

The estate is filed as a ten-year series — a Pilot Issue that begins an archival horizon that runs from 2026 to 2036, one issue per year, each issue carrying the WORM head **E965B01D** and the commitment to update every number the previous issue filed. The ten-year horizon is the estate's refusal to be a thing that publishes once and disappears. The horizon is the thing that makes the estate a serial, and a serial is the thing that lets a reader, ten years from now, trace any claim back to the issue that first made it, verify the claim against the chain, and check whether the claim was updated or withdrawn. The ten-year horizon is the WORM ledger applied to time. The ledger is the receipt. The receipt is the series. The series is the horizon.

11.4 Issue 2 — "Flow"

Issue 2 of Vaguely is titled *Flow*, and the title is the thing the Pilot Issue points toward. If the Pilot is the receipt, Flow is the validation. If the Pilot is the architecture, Flow is the deployment. If the Pilot is the CANDIDATE, Flow is the VERIFIED — the issue that

carries the results of the 72-hour protocol, the issue that files the DWS engineers' verdict, the issue that either elevates the status to Tier 3 or returns the primitive to the bench honestly. Flow is the issue the estate will demand more of than the Pilot, because Flow is the issue that carries the weight of the world's verdict. The Pilot is the ask. Flow is the answer. The ten-year horizon is the thing that holds the two apart.

The Estate Statement

Section 12 · the closing — One Primitive, Immutable Evidence, Honest Limits. The fall is part of the design. The design continues.

SECTION 12 OF 12 · THE ESTATE STATEMENT

12.1 One Primitive

The claim is one master primitive, reconfigured across water, phone, and vehicle — the same four layers, the same six primitives, the same three levels. The claim is the thing the Skylantern Evidence Architecture files, and the filing is the thing the four-layer stack makes true: the primitive is a primitive because it is small enough to print, disciplined enough to trust, and honest enough to file against itself. The claim is not a marketing claim. The claim is a structural claim, and the structure is the thing the next two clauses hold.

12.2 Immutable Evidence

The mechanism is the WORM chain, the Ed25519 signature, the SHA-256 hash, the chain head **E965B01D** — the thing that proves the record was not altered. The mechanism is the thing the CircuitBreaker kernel enforces: append-only, no edits, no deletes, no reordering. The mechanism is the thing the 11-Beat Loop feeds: signed payloads, folded into Merkle roots, submitted to the kernel, anchored to the head. The mechanism is the receipt. The receipt is the thing a partner checks against the chain. The chain is the thing the WORM head anchors.

12.3 Honest Limits

The limit is the Zero Fabrication Rule, the key rule, the negative scopes, the refusal to actuate, the refusal to fabricate, the refusal to exceed the R9-to-R9,999 band. The limit is the Oracle Boundary: the chain proves the record was not altered, but it does not prove the reading was true. The limit is the CANDIDATE state: a single witness is not a fact. The limit is the Tier 2 status: the estate is specified, not verified, and the verification is the thing the DWS engineers are asked to provide. The limit is the estate's confession of what it can do and what it refuses to claim.

12.4 The Fall Is Part of the Design

[SEALED] VVU · ESTATE STATEMENT

One Primitive. Immutable Evidence. Honest Limits. Governed by the Zero Fabrication Rule: nothing is presented as operational unless it is. Every claim carries a tier tag. We prove the record, not the reality. Status: Tier 2 (Specified). The fall is part of the design.

This document is the comprehensive planning framework the founder hands to every partner. It compiles the estate — the architecture, the testimony, the kernel, the water

hub, the audit, the platform, the go-live, the twelve clusters, the evidence architecture, and the two codebases — into a single, navigable thing a partner can read, verify, and act on. The platform is live. The clusters are drafted. The articles are hosted. The pack is filed. The receipt is sealed. The 72-hour protocol is the thing the DWS is asked to run. The Kit of Parts is the thing the Quad-Partite Alliance is asked to sponsor. The site and the support are the thing the municipality is asked to provide. The acceleration is the thing Propella and PIA are asked to fund. The estate has filed all of it, in writing, before the answer, with the radical honesty the estate was built to enforce. **The fall is part of the design.** The design continues.